

FINAL PROJECT REPORT

Heart Disease Prediction Using Machine Learning

Course: CBIO313 – Data Mining and Machine Learning

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BRFSS 2015 Health Indicators Dataset

1. Abstract

This project develops a machine-learning system for predicting whether an individual has a history of heart disease or heart attack using health and demographic indicators from the 2015 Behavioral Risk Factor Surveillance System (BRFSS). The workflow uses a large public dataset, performs exploratory data analysis and preprocessing, creates additional health-risk features, applies statistical feature selection, and compares several classification algorithms. Logistic Regression, Random Forest, and Gradient Boosting were evaluated, followed by GridSearchCV tuning of Logistic Regression. The tuned Logistic Regression model achieved a ROC-AUC of 0.8388, recall of 0.7962, precision of 0.2387, and F1-score of 0.3672 on the held-out test set. The model was serialized with joblib and integrated into a Streamlit web application that provides an interactive risk-assessment interface.

2. Introduction

Cardiovascular disease is strongly associated with measurable health, lifestyle, and demographic factors such as high blood pressure, high cholesterol, smoking, diabetes, physical inactivity, general health, and age. Machine learning can be used to identify patterns in these variables and produce a predictive classification model. The objective of this project is to build an end-to-end pipeline that predicts the target variable *HeartDiseaseorAttack* and demonstrates how the resulting model can be deployed as a user-facing application.

3. Problem Statement and Objectives

Problem: Can a person's health indicators be used to predict whether they have experienced heart disease or a heart attack?

Objectives:

- Prepare and inspect the BRFSS 2015 health-indicator dataset.
- Explore relationships between health indicators and the target variable.
- Engineer meaningful cardiovascular-risk features.
- Select informative predictors using an embedded/statistical feature-selection workflow.
- Train and compare multiple classification algorithms.
- Tune the strongest candidate using cross-validation.
- Evaluate the final model using precision, recall, F1-score and ROC-AUC.
- Deploy the trained model in a Streamlit web application.

4. Dataset

The repository contains the BRFSS 2015 Heart Disease Health Indicators dataset with **253,680** observations and 22 original variables. The notebook selects 11 columns for the initial modelling dataframe: the target plus ten health/demographic predictors. The target is *HeartDiseaseorAttack*, where 1 indicates a reported history of heart disease or heart attack and 0 indicates no such history.

Item	Description
Source	BRFSS 2015 Heart Disease Health Indicators dataset
Records	253,680
Original variables	22
Target	HeartDiseaseorAttack
Target = 0	229,787
Target = 1	23,893

Selected initial predictors	HighBP, HighChol, BMI, Smoker, Diabetes, PhysActivity, GenHlth, Sex, Age, Income
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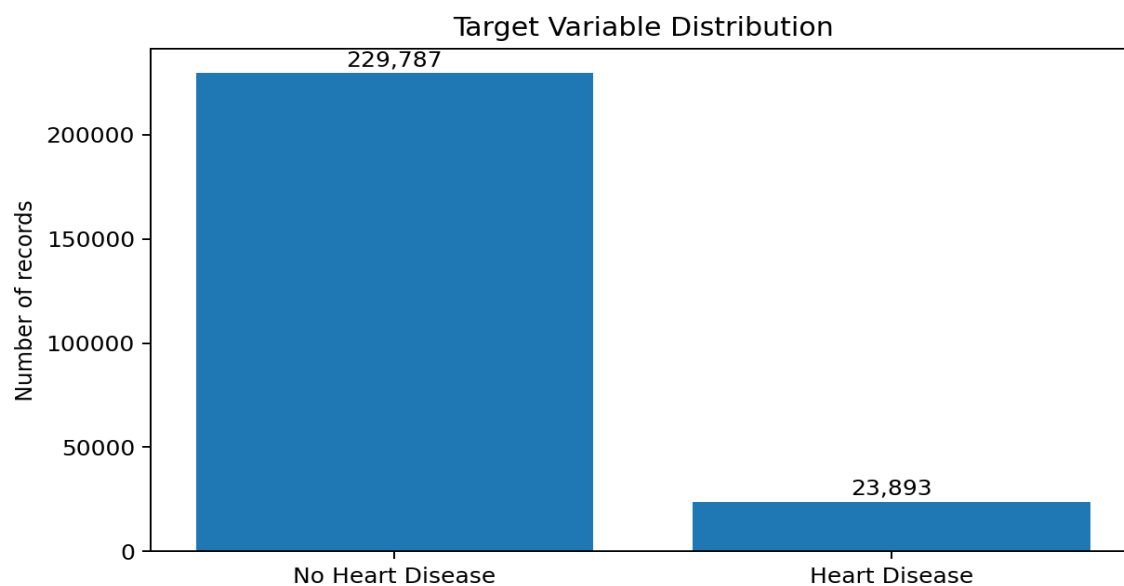


Figure 1. The target is imbalanced, with substantially fewer positive heart-disease cases.

5. Data Preprocessing and Cleaning

The notebook checks data types and missing values before modelling. The selected modelling data contain only two missing values, which are removed before feature selection. BMI contains unusually high values; the notebook caps BMI values above 80 to reduce the influence of extreme data-entry values.

The target distribution is imbalanced: approximately 9.4% of observations are positive cases. Because missing the positive class is important in a health-risk prediction setting, the Logistic Regression models use `class_weight='balanced'`. Stratified train-test splitting is also used to preserve the class distribution.

6. Exploratory Data Analysis

The analysis includes univariate and bivariate visualizations covering target distribution, BMI distributions, correlations, and risk-factor prevalence. The plots indicate clear differences in several health indicators between people with and without reported heart disease.

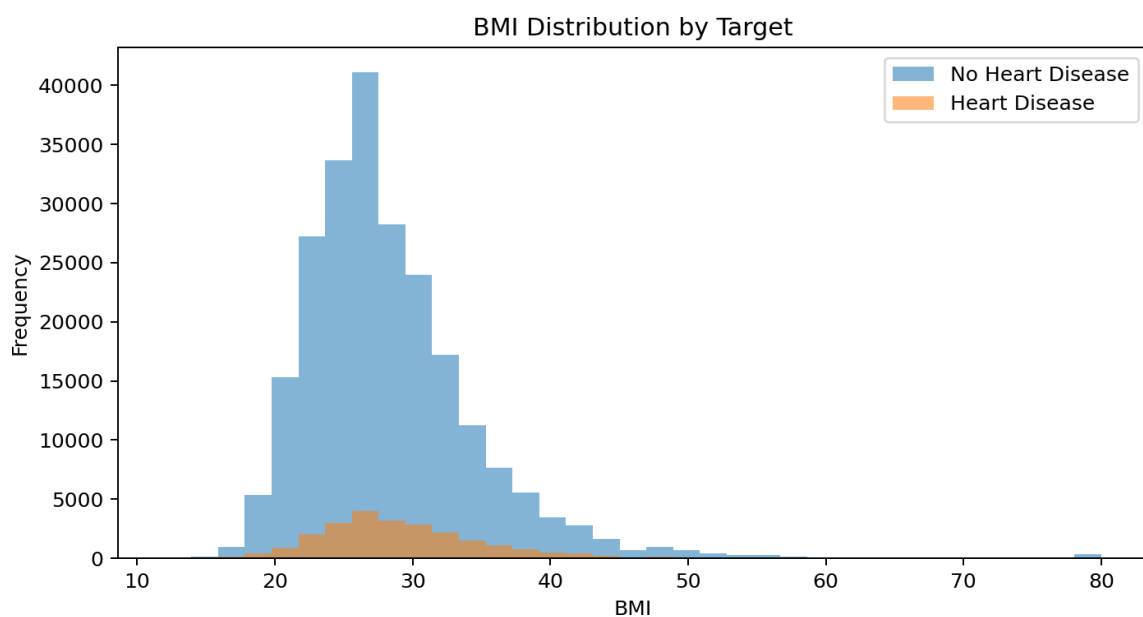


Figure 2. BMI distributions after capping extreme values at 80.

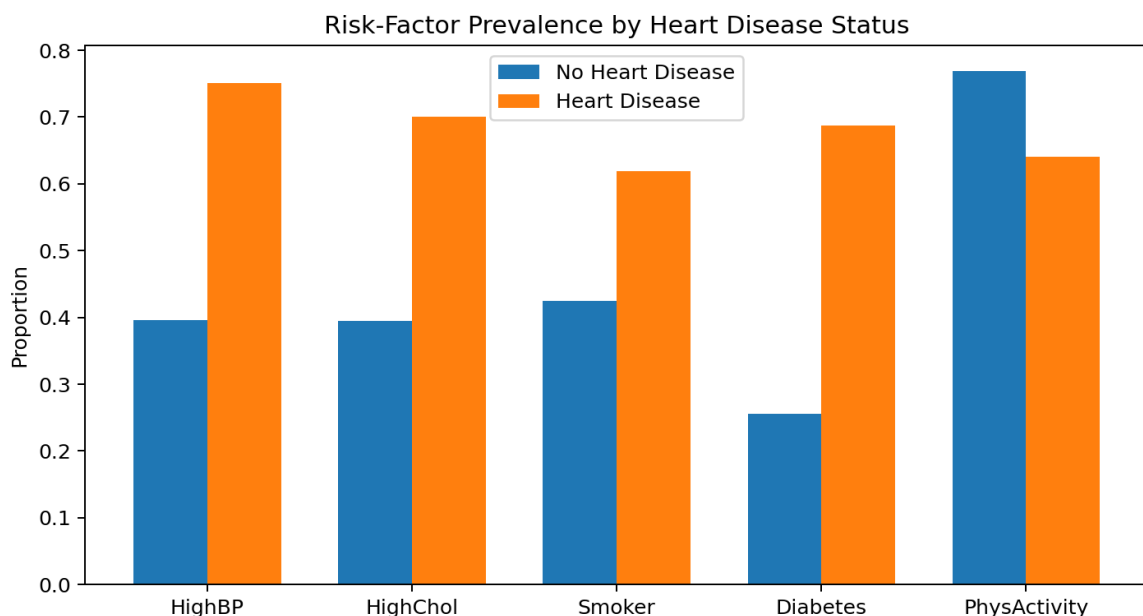


Figure 3. Risk-factor prevalence differs between the two target classes.

7. Feature Engineering

Two derived variables are created in the notebook:

Health_Risk_Score: combines HighBP, HighChol, Smoker, Diabetes, and lack of physical activity into a single risk indicator. Higher values represent a greater accumulation of baseline risk factors.

BMI_Category: converts continuous BMI into four ordinal groups: underweight, normal weight, overweight, and obese.

8. Feature Selection

The notebook uses SelectKBest with the ANOVA F-test to rank candidate variables by their association with the target. The highest-scoring variables are Health_Risk_Score, GenHlth, Age, HighBP, HighChol, Diabetes, Income, Smoker, PhysActivity, and Sex. These ten variables are used for the modelling stage.

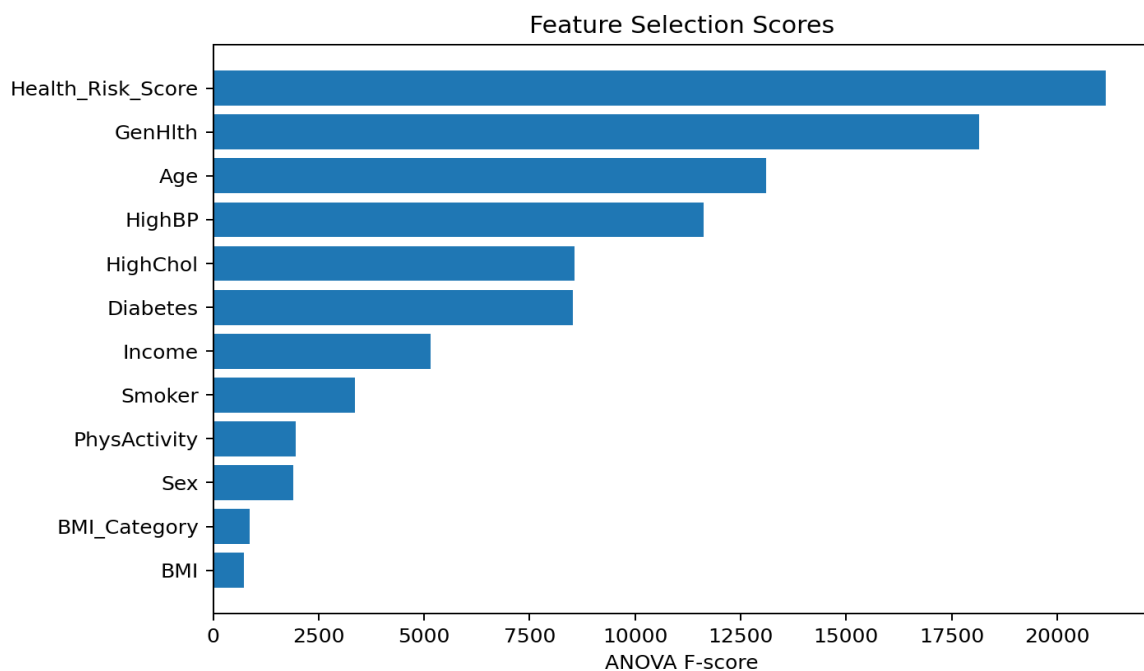


Figure 4. ANOVA F-scores used to rank candidate predictors.

9. Model Development

The data are split into 80% training and 20% testing using stratification and random_state=42. StandardScaler is fitted on the training set for the Logistic Regression models. Three algorithms are compared: Logistic Regression with class balancing, Random Forest with class balancing, and Gradient Boosting. GridSearchCV with three-fold cross-validation is then used to tune Logistic Regression using ROC-AUC as the scoring metric.

10. Results

Model	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.2387	0.7962	0.3672	0.8388
Random Forest	0.2394	0.5844	0.3397	0.7518
Gradient Boosting	0.5225	0.0607	0.1087	0.8412
Tuned Logistic Regression	0.2387	0.7962	0.3672	0.8388

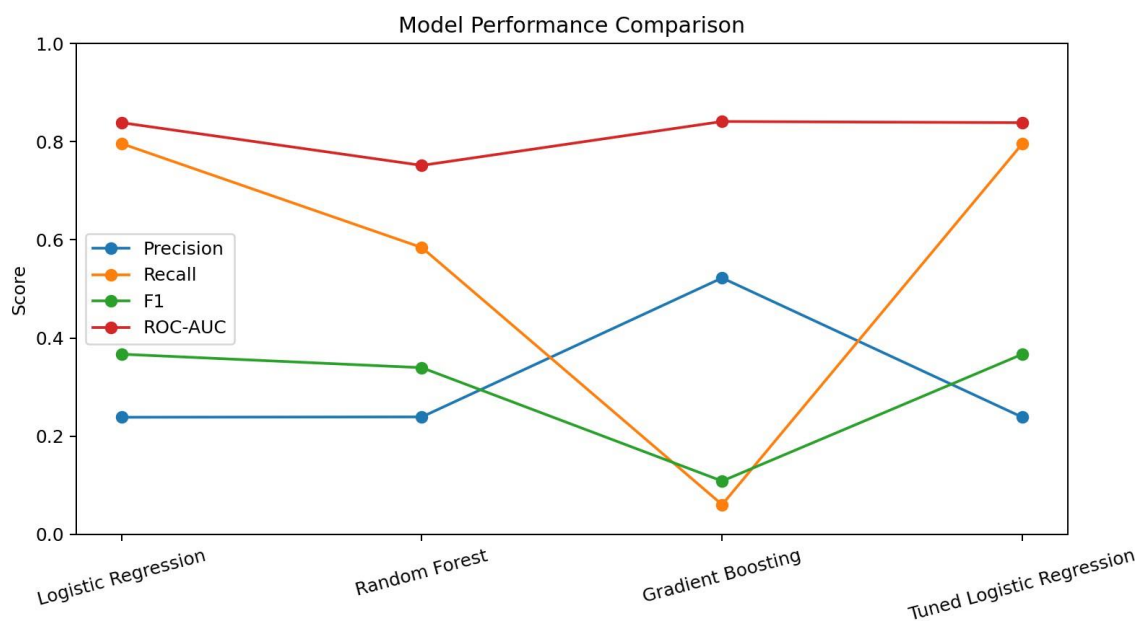


Figure 5. Comparison of the reported test-set metrics.

11. Best Model and Interpretation

Based on the notebook's evaluation, the tuned Logistic Regression model is retained as the final model. GridSearchCV selected $C=10$ and solver='liblinear'. On the test set, the model achieved ROC-AUC 0.8388, recall 0.7962, precision 0.2387, and F1-score 0.3672. Its recall is relatively high, which is useful for identifying a larger proportion of positive cases in this imbalanced dataset. The lower precision means that a meaningful number of predicted positive cases are false positives; therefore, the output should be treated as a risk-screening aid rather than a clinical diagnosis.

The classification report recorded 45,957 negative and 4,779 positive test observations. For the positive class, recall was 0.80 and F1-score was 0.37. Overall accuracy was approximately 0.74. Because accuracy alone can be misleading under class imbalance, ROC-AUC, precision, recall and F1-score are emphasized.

12. Deployment

The trained estimator, scaler, and selected feature list are serialized with joblib and stored in the repository. The Streamlit application loads these artifacts and provides an interactive form for demographic, clinical, health-history, and lifestyle inputs. It displays a predicted class and model probability together with a list of active risk factors.

The repository README also documents the deployed Streamlit application and the project structure. The application is intended as an educational demonstration of machine-learning deployment and not as a substitute for professional medical assessment.

13. Limitations

The dataset is observational and reflects self-reported health information, so associations should not be interpreted as causal relationships. The positive class is substantially smaller than the negative class, which affects precision and other metrics. The model also depends on the feature definitions and coding scheme used by BRFSS. Finally, predictive performance on this dataset does not establish clinical validity in other populations or settings.

14. Conclusion

This project demonstrates a complete machine-learning workflow for heart-disease risk classification: dataset inspection, preprocessing, exploratory analysis, feature engineering, feature selection, model comparison, hyperparameter tuning, evaluation, serialization, and deployment. The final tuned Logistic Regression model provides a useful balance for this project, particularly in terms of recall and ROC-AUC. The resulting Streamlit application translates the trained model into an accessible interactive interface.

15. Future Work

- Evaluate additional models and calibrated probability outputs.
- Experiment with threshold selection to explicitly trade off recall and precision.
- Use external validation data to assess generalization.
- Perform more extensive fairness and subgroup analysis.
- Add clearer clinical disclaimers and input validation to the deployed interface.
- Reconcile and test the exact preprocessing pipeline between notebook training and the Streamlit application before production use.

16. References and Project Resources

- BRFSS 2015 Heart Disease Health Indicators dataset (included in the repository).

- Scikit-learn documentation and algorithms used in the notebook.
- Streamlit documentation for the deployed application.
- GitHub repository: [Farah-ashraf600/MLPROJ](https://github.com/Farah-ashraf600/MLPROJ).